COMP3211 Software Engineering
Assignment 2

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1 Class Diagram

The class diagram is shown in Figure 1.

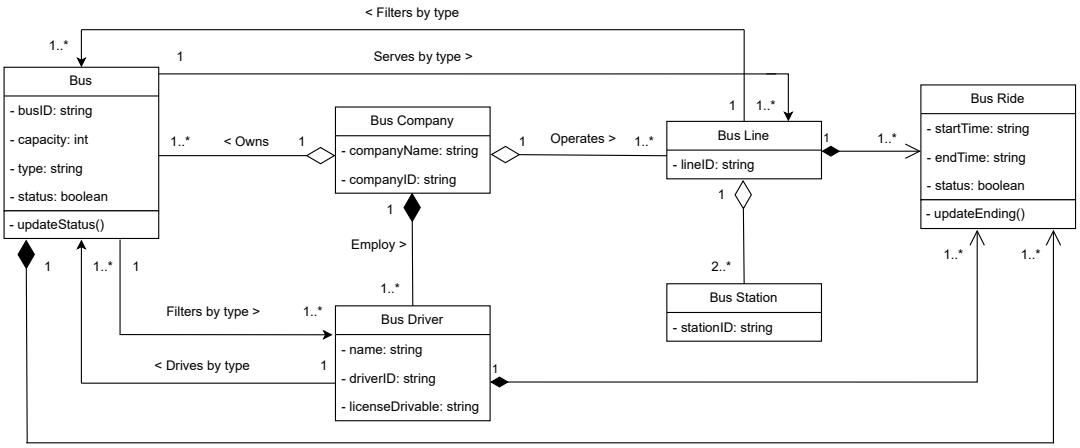

Figure 1: Class Diagram

2 Architectural Design

Advantage of Design 1 Integrating the presentation and application layer into the client side can improve the performance of the system by reducing the server's load and minimizing the negative impacts on processing procedures due to unstable network connections.

Advantage of Design 2 The centralized functions of application processing can improve the system's security and maintainability, as malicious users are prevented from processing procedures

which are not located on the client side, and evolution on the application processing can be deployed on the server side without the need to update the clients.

3 Software Testing

The set of unit tests is given in Table 1.

x	y	z	ExpectedResult	BranchCovered
-1	1	1	false	c2==true
1	-1	1	false	c2==true
1	1	3	false	c2==false, c5==true
1	2	10	false	c2==false, c5==true
1	3	1	false	c2==false, c5==false, c7==true
1	10	2	false	c2==false, c5==false, c7==true
3	1	1	false	c2==false, c5==false, c7==false, c9==true
10	2	1	false	c2==false, c5==false, c7==false, c9==true
1	3	3	true	c2==false, c5==false, c7==false, c9==false
3	4	5	true	c2==false, c5==false, c7==false, c9==false

Table 1: Unit Tests

4 Software Maintenance

(1) Corrective Maintenance: the reactive correction of existing bugs according to user reports; **(2)** Adaptive Maintenance: the reactive enhancement of the software to adapt to possible changes in the environment; **(3)** Preventive Maintenance: the proactive enhancement on the software to prevent potential software defects; **(4)** Perfective Maintenance: the proactive improvement of the software functionality or performance to enhance the user experience.

The four types of maintenance are difficult to distinguish in practice, as **(1)** they are often intertwined to perform simultaneously throughout the process of software maintenance; **(2)** their outcomes can be overlapping, as adapting to changes in the environment (adaptive) and correction of current bugs (corrective) might also prevent potential defects (preventive) and influence the software functionality or performance (perfective), and vice verssa; **(3)** the classification of maintenance types is subjective to the motivation of maintaining, while the contents of different types of maintenance can be similar or even identical.